

lyche-numerical-linear-algebra: residuals

equations measured 2026-09-05T12:57:29Z against the model of 2026-09-01

Corrected (8)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0326 (was)	133	0.118	<pre>\begin{aligned} & \boldsymbol{Q}_1:=\left[\right. \\ \begin{array}{lllll} \boldsymbol{q}_1, & \ldots, & \boldsymbol{q}_n \end{array} \\ & \left. \right], \quad \& \\ \boldsymbol{R}_1:= & \& \boldsymbol{q}_j:=\frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j=1,\ldots,n \text{ and } \\ \left[\begin{array}{cccccc} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \boldsymbol{a}_n^T \boldsymbol{q}_3 & \\ & & & \ddots & \vdots & \vdots \\ & & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{array} \right]. \end{aligned}</pre>	$\boldsymbol{Q}_1 := [\boldsymbol{q}_1, \dots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and}$ $\boldsymbol{R}_1 := \left[\begin{array}{cccccc} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \boldsymbol{a}_n^T \boldsymbol{q}_3 & \\ & & & \ddots & \vdots & \vdots \\ & & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{array} \right].$	
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- E00326 (now)	133	—	$\begin{aligned} & \boldsymbol{Q}_1 := [\boldsymbol{q}_1, \dots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and} \\ & \boldsymbol{R}_1 := \begin{bmatrix} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots & \vdots \\ & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{bmatrix} \end{aligned}$	$\boldsymbol{Q}_1 := [\boldsymbol{q}_1, \dots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and}$ $\boldsymbol{R}_1 := \begin{bmatrix} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots & \vdots \\ & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{bmatrix}.$	
basis: inferred / ink					
lyche-numerical-linear-algebra- E00369 (was)	150	■0.298	$\boldsymbol{J} := \operatorname{diag}(\boldsymbol{U}_1, \boldsymbol{U}_2) = \begin{bmatrix} 2 & 1 & 0 & & & & & \\ 0 & 2 & 1 & & & & & \\ 0 & 0 & 2 & & 1 & & & \\ & & 0 & 1 & & & & \\ & & 0 & 2 & & & & \\ & & & & 0 & & & \\ & & & & & 0 & 1 & \end{bmatrix} \in \mathbb{R}^{8 \times 8}.$	$\boldsymbol{J} := \operatorname{diag}(\boldsymbol{U}_1, \boldsymbol{U}_2) = \begin{bmatrix} 2 & 1 & 0 & & & & & \\ 0 & 2 & 1 & & & & & \\ 0 & 0 & 2 & & & & & \\ & & 2 & 1 & & & & \\ & & 0 & 2 & & & & \\ & & & 2 & 1 & & & \\ & & & & 3 & 1 & & \\ & & & & 0 & 3 & & \end{bmatrix} \in \mathbb{R}^{8 \times 8}.$	
lyche-numerical-linear-algebra- E00369 (now)	150	—	$\begin{aligned} & d_i \text{ for } i = 1, \dots, n, \quad b_{i+1,i} := a_i, \quad b_{i,i+1} := c_i \text{ for } i \text{ otherwise, } \in \mathbb{C}^{m \times n} \text{ and } 1 \leq i_1 < i_2 < \dots < i_r \leq m, 1 \leq j \leq n \\ & \text{matrix } A(i, j) \in \mathbb{C}^{r \times c} \text{ is the submatrix of } A \end{aligned}$	$d_i \text{ for } i = 1, \dots, n, b_{i+1,i} := a_i, b_{i,i+1} := c_i \text{ for } i \text{ otherwise, } \in \mathbb{C}^{m \times n} \text{ and } 1 \leq i_1 < i_2 < \dots < i_r \leq m, 1 \leq j \leq n \text{ matrix } A(i, j) \in \mathbb{C}^{r \times c} \text{ is the submatrix of } A$	$\boldsymbol{J} := \operatorname{diag}(\boldsymbol{U}_1, \boldsymbol{U}_2) = \begin{bmatrix} 2 & 1 & 0 & & & & & \\ 0 & 2 & 1 & & & & & \\ 0 & 0 & 2 & & & & & \\ & & 2 & 1 & & & & \\ & & 0 & 2 & & & & \\ & & & 2 & 1 & & & \\ & & & & 3 & 1 & & \\ & & & & 0 & 3 & & \end{bmatrix} \in \mathbb{R}^{8 \times 8}.$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra E08579 (was)	215	■ 0.322	$\begin{aligned} & \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array} \right] \left[\begin{array}{c} 1 \\ \vdots \\ 1 \\ m \end{array} \right] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} m & \sum t_k \\ \sum t_k & \sum t_k^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \\ & = \left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array} \right] \left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array} \right] = \begin{bmatrix} \sum y_k \\ \sum t_k y_k \end{bmatrix} = \boldsymbol{A}^* \boldsymbol{b}, \end{aligned}$	$\boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array} \right] \left[\begin{array}{c} 1 \\ \vdots \\ 1 \\ m \end{array} \right] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} m & \sum t_k \\ \sum t_k & \sum t_k^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$ $= \left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array} \right] \left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array} \right] = \begin{bmatrix} \sum y_k \\ \sum t_k y_k \end{bmatrix} = \boldsymbol{A}^* \boldsymbol{b},$	
lyche-numerical-linear-algebra E08579 (now)	215	—	$\text{\text{nal if } } a_{ij} = 0 \text{\text{ for } } i \neq j. \text{\text{ }} \text{\text{triangular or \textbf{right triangular} if } } a_{ij} = 0 \text{\text{ for } } i > j. \text{\text{ }} \text{\text{triangular or \textbf{left triangular} if } } a_{ij} = 0 \text{\text{ for } } i < j. \text{\text{ }} \text{\textbf{Hessenberg if } } a_{ij} = 0 \text{\text{ for } } i > j + 1. \text{\text{ }} \text{\textbf{Hessenberg if } } a_{ij} = 0 \text{\text{ for } } i < j + 1. \text{\text{ }} \text{\text{gonal if } } a_{ij} = 0 \text{\text{ for } } i - j > 1. \text{\text{ }} \text{\text{ded if } } a_{ij} = 0 \text{\text{ for } } i - j > d. \text{\text{[lex] \text{the following notations for diagonal- and tridiagonal } } n \times n$	<i>(not rendered)</i>	$\boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array} \right] \left[\begin{array}{c} 1 \\ \vdots \\ 1 \\ t_m \end{array} \right] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} m & \sum t_k \\ \sum t_k & \sum t_k^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$ $= \left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array} \right] \left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array} \right] = \begin{bmatrix} \sum y_k \\ \sum t_k y_k \end{bmatrix} = \boldsymbol{A}^* \boldsymbol{b},$

basis: inferred / ink

Conf.

MathPix confidence:

■

≥0.9

■

0.5-0.9

■

<0.5

— no value

Residual

render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- E09630 (was)	229	0.167	<pre>\begin{aligned} & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array}\right] \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{array}\right] = \left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array}\right] = \left[\begin{array}{c} \alpha_i \boldsymbol{u}_i \\ \alpha_i \boldsymbol{v}_i \end{array}\right] = \alpha_i \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{array}\right], \quad i = 1, \dots, r, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array}\right] \left[\begin{array}{c} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{array}\right] = \left[\begin{array}{c} -\boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array}\right] = \left[\begin{array}{c} -\alpha_i \boldsymbol{u}_i \\ \alpha_i \boldsymbol{v}_i \end{array}\right] = -\alpha_i \left[\begin{array}{c} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{array}\right], \quad i = 1, \dots, r, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array}\right] \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{0} \end{array}\right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array}\right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{0} \end{array}\right] = 0 \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{0} \end{array}\right], \quad i = r + 1, \dots, m, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array}\right] \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{v}_i \end{array}\right] = \left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{0} \end{array}\right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{0} \end{array}\right] = 0 \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{v}_i \end{array}\right], \quad i = r + 1, \dots, n. \end{aligned}</pre>	$\begin{bmatrix} \mathbf{0} & \mathbf{A} \\ \mathbf{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{u}_i \\ \mathbf{v}_i \end{bmatrix} = \begin{bmatrix} \mathbf{A} \mathbf{v}_i \\ \mathbf{A}^* \mathbf{u}_i \end{bmatrix} = \begin{bmatrix} \alpha_i \mathbf{u}_i \\ \alpha_i \mathbf{v}_i \end{bmatrix} = \alpha_i \begin{bmatrix} \mathbf{u}_i \\ \mathbf{v}_i \end{bmatrix}, \quad i = 1, \dots, r,$ $\begin{bmatrix} \mathbf{0} & \mathbf{A} \\ \mathbf{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{u}_i \\ -\mathbf{v}_i \end{bmatrix} = \begin{bmatrix} -\mathbf{A} \mathbf{v}_i \\ \mathbf{A}^* \mathbf{u}_i \end{bmatrix} = \begin{bmatrix} -\alpha_i \mathbf{u}_i \\ \alpha_i \mathbf{v}_i \end{bmatrix} = -\alpha_i \begin{bmatrix} \mathbf{u}_i \\ -\mathbf{v}_i \end{bmatrix}, \quad i = 1, \dots, r,$ $\begin{bmatrix} \mathbf{0} & \mathbf{A} \\ \mathbf{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{u}_i \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{A}^* \mathbf{u}_i \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} = 0 \begin{bmatrix} \mathbf{u}_i \\ \mathbf{0} \end{bmatrix}, \quad i = r + 1, \dots, m,$ $\begin{bmatrix} \mathbf{0} & \mathbf{A} \\ \mathbf{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ \mathbf{v}_i \end{bmatrix} = \begin{bmatrix} \mathbf{A} \mathbf{v}_i \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} = 0 \begin{bmatrix} \mathbf{0} \\ \mathbf{v}_i \end{bmatrix}, \quad i = r + 1, \dots, n.$	
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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lyche-numerical-linear-algebra-E09630 (now)	229	—	$\begin{aligned} &a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ &a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ &\qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \\ &a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{aligned}$	$\begin{aligned} \begin{bmatrix} \mathbf{0} & A \\ A^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} u_i \\ v_i \end{bmatrix} &= \begin{bmatrix} A v_i \\ A^* u_i \end{bmatrix} = \begin{bmatrix} \alpha_i u_i \\ \alpha_i v_i \end{bmatrix} = \alpha_i \begin{bmatrix} u_i \\ v_i \end{bmatrix}, \quad i = 1, \dots, r, \\ \begin{bmatrix} \mathbf{0} & A \\ A^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} u_i \\ -v_i \end{bmatrix} &= \begin{bmatrix} -A v_i \\ A^* u_i \end{bmatrix} = \begin{bmatrix} -\alpha_i u_i \\ \alpha_i v_i \end{bmatrix} = -\alpha_i \begin{bmatrix} u_i \\ -v_i \end{bmatrix}, \quad i = 1, \dots, r, \\ \begin{bmatrix} \mathbf{0} & A \\ A^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} u_i \\ \mathbf{0} \end{bmatrix} &= \begin{bmatrix} \mathbf{0} \\ A^* u_i \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} = 0 \begin{bmatrix} u_i \\ \mathbf{0} \end{bmatrix}, \quad i = r + 1, \dots, m, \\ \begin{bmatrix} \mathbf{0} & A \\ A^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ v_i \end{bmatrix} &= \begin{bmatrix} A v_i \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} = 0 \begin{bmatrix} \mathbf{0} \\ v_i \end{bmatrix}, \quad i = r + 1, \dots, n. \end{aligned}$	
basis: inferred / ink					
lyche-numerical-linear-algebra-E09670 (was)	241	■ 0.390	$\begin{aligned} &a_{ii} = 2d, i=1, \ldots, n, \\ &a_{i,i+1} = a_{i+1,i} = a, \quad i=1, \ldots, n-1, \quad i \neq m, 2m, \ldots, (m-1)m, \\ &a_{i,i+m} = a_{i+m,i} = a, \quad i=1, \ldots, n-m, \\ &a_{ij} = 0, \text{ otherwise} \end{aligned}$	$\begin{aligned} &a_{ii} = 2d, i = 1, \dots, n, \\ &a_{i,i+1} = a_{i+1,i} = a, \quad i = 1, \dots, n-1, \quad i \neq m, 2m, \dots, (m-1)m, \\ &a_{i,i+m} = a_{i+m,i} = a, \quad i = 1, \dots, n-m, \\ &a_{ij} = 0, \text{ otherwise} \end{aligned}$	$\begin{aligned} &a_{ii} = 2d, i = 1, \dots, n, \\ &a_{i,i+1} = a_{i+1,i} = a, \quad i = 1, \dots, n-1, \quad i \neq m, 2m, \dots, (m-1)m, \\ &a_{i,i+m} = a_{i+m,i} = a, \quad i = 1, \dots, n-m, \\ &a_{ij} = 0, \text{ otherwise}, \end{aligned}$
lyche-numerical-linear-algebra-E09670 (now)	241	—	$\begin{bmatrix} \vdots & \vdots & \cdots & \vdots \\ a_{r_2, c_1} & a_{r_2, c_1+1} & \cdots & a_{r_2, c_2} \end{bmatrix}$	$\begin{bmatrix} \vdots & \vdots & \cdots & \vdots \\ a_{r_2, c_1} & a_{r_2, c_1+1} & \cdots & a_{r_2, c_2} \end{bmatrix}$	$\begin{aligned} &a_{ii} = 2d, i = 1, \dots, n, \\ &a_{i,i+1} = a_{i+1,i} = a, \quad i = 1, \dots, n-1, \quad i \neq m, 2m, \dots, (m-1)m, \\ &a_{i,i+m} = a_{i+m,i} = a, \quad i = 1, \dots, n-m, \\ &a_{ij} = 0, \text{ otherwise}, \end{aligned}$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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lyche-numerical-linear-algebra-EQ0729 (was)	258	■ 0.002	$\begin{array}{lll} \left(\boldsymbol{F}_{2m}\boldsymbol{P}_{2m}\right)_{p,q} & = \omega_{2m}^{j(2k)} & = \omega_m^{jk}, \\ \left(\boldsymbol{F}_{2m}\boldsymbol{P}_{2m}\right)_{p+m,q} & = \omega_{2m}^{(j+m)(2k)} & = \omega_m^{(j+m)k} = \omega_m^{jk}, \\ \left(\boldsymbol{F}_{2m}\boldsymbol{P}_{2m}\right)_{p,q+m} & = \omega_{2m}^{j(2k+1)} & = \omega_{2m}^j \omega_m^{jk}, \\ \left(\boldsymbol{F}_{2m}\boldsymbol{P}_{2m}\right)_{p+m,q+m} & = \omega_{2m}^{(j+m)(2k+1)} & = \omega_{2m}^{j+m} \omega_m^{(j+m)k} = \omega_{2m}^j \omega_m^{jk}. \end{array}$	$(j+m)$	
lyche-numerical-linear-algebra-EQ0729 (now)	258	—	$(DX + XD)_{j,k} = \sum_{\ell=1}^m d_{j,\ell} x_{\ell,k} + \sum_{\ell=1}^m x_{j,\ell} d_{\ell,k} = \lambda_j x_{j,k} + \lambda_k x_{j,k} \quad \text{for all } j, k$		
basis: inferred / ink					
lyche-numerical-linear-algebra-EQ0763 (was)	271	■ 0.402	$\boldsymbol{G}_{\omega} = \begin{bmatrix} l & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{bmatrix}.$		
lyche-numerical-linear-algebra-EQ0763 (now)	271	—	$\langle j_1, j_2 \mid j_c \rangle \quad \begin{bmatrix} \cdot & \cdot & \cdots & \cdot \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}$		
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- EQ0855	296	■ 0.386 ● C -60	$\begin{array}{l} \boldsymbol{t}_0 = \begin{bmatrix} 3 \\ -3/2 \end{bmatrix}, \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \begin{bmatrix} -1/4 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{r}_1 = \begin{bmatrix} 0 \\ 3/4 \end{bmatrix}, \quad \beta_0 = 1/4, \\ \boldsymbol{p}_1 = \begin{bmatrix} 3/8 \\ 3/4 \end{bmatrix}, \quad \boldsymbol{t}_1 = \begin{bmatrix} 0 \\ 9/8 \end{bmatrix}, \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \mathbf{0}, \quad \boldsymbol{r}_2 = \mathbf{0}. \end{array}$	$\boldsymbol{t}_0 = \begin{bmatrix} 3 \\ -3/2 \end{bmatrix}, \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \begin{bmatrix} -1/4 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{r}_1 = \begin{bmatrix} 0 \\ 3/4 \end{bmatrix}, \quad \beta_0 = 1/4, \\ \boldsymbol{p}_1 = \begin{bmatrix} 3/8 \\ 3/4 \end{bmatrix}, \quad \boldsymbol{t}_1 = \begin{bmatrix} 0 \\ 9/8 \end{bmatrix}, \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \mathbf{0}, \quad \boldsymbol{r}_2 = \mathbf{0}.$	
lyche-numerical-linear-algebra- EQ1043	353	■ 0.971 ● C +78	$\begin{array}{l} \boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{12}^*} \begin{bmatrix} x & x & x & x \\ \mathbf{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{23}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{34}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{bmatrix}. \end{array}$	<i>(not rendered)</i>	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{12}^*} \begin{bmatrix} x & x & x & x \\ \mathbf{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{23}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{34}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0195 (3.9)	82	■ 1.000 ● C +10	$\begin{array}{l} \text{Listing 3.3 cforwardsolve} \\ N_{LU} := \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \end{array}$	<i>(not rendered)</i>	$N_{LU} := \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n$
lyche-numerical-linear-algebra- EQ0232 (3.22)	93	■ 1.000 ● C +54	$\begin{array}{l} \text{Listing 3.5 cbacksolve} \\ \boldsymbol{A}((k+1):n, (k+1):n) \boldsymbol{b}_k((k+1):n) = -\boldsymbol{A}((k+1):n, k) \boldsymbol{b}_k(k), \quad k = 1, \dots, n-1. \end{array}$	<i>(not rendered)</i>	$\boldsymbol{A}((k+1):n, (k+1):n) \boldsymbol{b}_k((k+1):n) = -\boldsymbol{A}((k+1):n, k) \boldsymbol{b}_k(k), \quad k = 1, \dots, n-1.$
lyche-numerical-linear-algebra- EQ0258 (4.7)	106	■ 1.000 ● C +54	$\begin{array}{l} \text{Listing 4.2 bandcholesky} \\ 0 < (\leq) (\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j)^* \boldsymbol{A} (\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j) = \alpha ^2 a_{ii} + \beta ^2 a_{jj} + 2 \operatorname{Re} (\bar{\alpha} \beta a_{ij}). \end{array}$	<i>(not rendered)</i>	$0 < (\leq) (\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j)^* \boldsymbol{A} (\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j) = \alpha ^2 a_{ii} + \beta ^2 a_{jj} + 2 \operatorname{Re} (\bar{\alpha} \beta a_{ij}).$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Flagged, not acted on

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-E00485	191	0.917 C +32	$\begin{aligned} & \ \boldsymbol{A} \boldsymbol{B}\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{B} \boldsymbol{x}\ } \frac{\ \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \\ & \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ \end{aligned}$	$\begin{aligned} \ \boldsymbol{A} \boldsymbol{B}\ &= \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{B} \boldsymbol{x}\ } \frac{\ \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \\ &\leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ . \end{aligned}$	
Conf. MathPix confidence: 0.917 0.5-0.9 <0.5 no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

194 more flagged rows below the band, stated as a count: 30 component, 164 weak; confidence high 138, mid 35, low 21, none 0